

PRACTICAL 1

AIM

To develop a Python program that creates a single player interactive game in which the user attempts to identify a randomly selected number generated by the computer.

THEORY

A single player game is a computer-based game that involves only one participant. In such games, the player interacts directly with the computer system, and the computer controls the game logic and responses.

In this program, Python is used to design a guessing game where the computer randomly selects a number within a given range. The player is required to guess this number through user input. The random library in Python is used to generate unpredictable numbers so that each game session is different.

The program makes use of decision-making statements such as *if-elif-else* to check whether the user's guess is correct, higher, or lower than the generated number. Based on this comparison, the program provides hints to guide the player toward the correct answer.

A loop structure is used to keep the game running until the player successfully guesses the correct number. Additionally, the program keeps track of the number of attempts taken by the player.

This game illustrates fundamental Python programming concepts including random number generation, control structures, loops, and user interaction, making it a useful example for understanding basic game logic implementation.

SOFTWARE USED

Jupyter Notebook

CODE

```
import random

def single_player_game():

    print("Welcome to the Number Guessing Game!")

    print("I have selected a number between 1 and 100.")

    print("Try to guess it.")


    number = random.randint(1, 100)

    attempts = 0
```

```
while True:

    guess = int(input("Enter your guess: "))

    attempts += 1

    if guess < number:

        print("Too low! Try again.")

    elif guess > number:

        print("Too high! Try again.")

    else:

        print("Congratulations! You guessed the correct number.")

        print("Total attempts:", attempts)

        break

single_player_game()
```

OUTPUT

```
*** Welcome to the Number Guessing Game!
I have selected a number between 1 and 100.
Try to guess it.
Enter your guess: 26
Too low! Try again.
Enter your guess: 87
Too high! Try again.
Enter your guess: 56
Too high! Try again.
Enter your guess: 52
Too high! Try again.
Enter your guess: 35
Too low! Try again.
Enter your guess: 47
Too high! Try again.
Enter your guess: 41
Too low! Try again.
Enter your guess: 40
Too low! Try again.
Enter your guess: 48
Too high! Try again.
Enter your guess: 46
Too high! Try again.
Enter your guess: 55
Too high! Try again.
Enter your guess: 42
Too low! Try again.
Enter your guess: 43
Too low! Try again.
Enter your guess: 44
Congratulations! You guessed the correct number.
Total attempts: 14
```